

Let's see what our thermal resistance would be:

$$Q = -kA \left(\frac{T_2 - T_1}{L} \right) = kA \left(\frac{T_1 - T_2}{L} \right) \Rightarrow \Delta T = T_1 - T_2$$

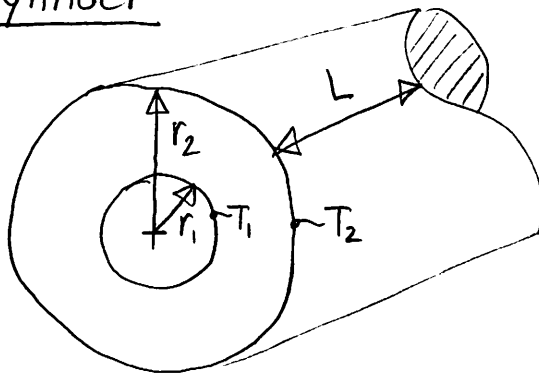
$$Q = kA \frac{\Delta T}{L}$$

But we know we want the form: $Q = \frac{\Delta T}{R}$

$$Q = kA \frac{\Delta T}{L} = \frac{\Delta T}{R}$$

$$\boxed{R = \frac{L}{kA}} \Rightarrow \text{Thermal resistance of a 1D slab}$$

② Cylinder



Writing out our heat equation in radial coordinates: (page 14)

$$\frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(k \frac{\partial T}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + Q''' = \rho c_p \frac{\partial T}{\partial t}$$

$T = f(r) \text{ only} \quad T = f(r) \text{ only} \quad Q''' = 0 \quad \text{Steady-state}$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) = 0$$

$$\frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) = 0 \quad (\text{integrate once})$$

$$\int \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) dr = \int 0 dr$$

$$r \frac{\partial T}{\partial r} = C_1 \quad (\text{Integrate once more})$$

$$\int \frac{\partial T}{\partial r} \partial r = \int \frac{C_1}{r} \partial r \quad * \text{ Remember } \int \frac{dr}{r} = \ln(r)$$

$$T = C_1 \ln r + C_2 \Rightarrow \text{Now we apply our boundary conditions}$$

$$T|_{r=r_1} = T_1 = C_1 \ln r_1 + C_2 \quad (1)$$

$$T|_{r=r_2} = T_2 = C_1 \ln r_2 + C_2 \quad (2)$$

Subtract (2) from (1)

$$T_1 - T_2 = C_1 \ln r_1 - C_1 \ln r_2 + \cancel{C_2} - \cancel{C_2}$$

$$T_1 - T_2 = C_1 (\ln r_1 - \ln r_2) \Rightarrow \text{Remember } \ln \text{ rules; } \begin{matrix} (1) \ln(a) - \ln(b) \\ = \ln(a/b) \end{matrix}$$

$$C_1 = \frac{T_1 - T_2}{\ln(r_1/r_2)} \quad (3)$$

Back substitute (3) into (1) & solve for C_2

$$C_2 = T_1 - \frac{(T_1 - T_2)}{\ln(r_1/r_2)} \ln(r_1) \quad (4)$$

Back substitute (3) & (4) into our solution for T above

$$T = \frac{T_1 - T_2}{\ln(r_1/r_2)} \ln r + T_1 - \frac{(T_1 - T_2) \ln r_1}{\ln(r_1/r_2)}$$

$$T - T_1 = (T_1 - T_2) \left[\frac{\ln r}{\ln(r_1/r_2)} - \frac{\ln(r_1)}{\ln(r_1/r_2)} \right] \Rightarrow \text{Remember the } \ln \text{ rules}$$

$$\frac{T_1 - T_1}{T_1 - T_2} = \frac{\ln(r/r_1)}{\ln(r_1/r_2)} \quad \text{or} \quad \boxed{\frac{T - T_1}{T_2 - T_1} = \frac{\ln(r/r_1)}{\ln(r_2/r_1)}} \Rightarrow \text{Radial temp. profile.}$$

Can we do the thermal resistance concept here too?

$$Q = \frac{\Delta T}{R} = -kA \left. \frac{\partial T}{\partial r} \right|_r = -kA \frac{\partial}{\partial r} \left[\frac{T_1 - T_2}{\ln(r_1/r_2)} \ln r + T_1 - \frac{(T_1 - T_2) \ln r_1}{\ln(r_1/r_2)} \right]$$

We know $\frac{\partial}{\partial r} [\ln r] = \frac{1}{r}$

$$\frac{\partial}{\partial r} (T_1) = 0 \quad \frac{\partial}{\partial r} (\dots) = 0$$

$$Q = -kA \frac{\partial}{\partial r} \left[\frac{T_1 - T_2}{\ln(r_1/r_2)} \ln r \right] = -kA \frac{(T_1 - T_2)}{\ln(r_1/r_2)} \cdot \frac{1}{r}$$

But $\ln(r_1/r_2) = -\ln(r_2/r_1) \Rightarrow$ Back substitute into above
 ↖ Length of tube (axial)

$$Q = kA \frac{T_1 - T_2}{\ln(r_2/r_1)} \cdot \frac{1}{r} \Rightarrow A = 2\pi r L$$

$$Q = 2\pi L k \frac{T_1 - T_2}{\ln(r_2/r_1)} \cdot \frac{1}{r} = \boxed{2\pi L k \frac{T_1 - T_2}{\ln(r_2/r_1)} = Q}$$

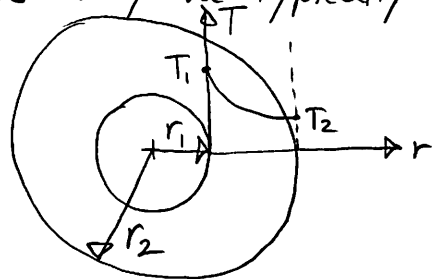
So for resistance, we want $R = \frac{\Delta T}{Q}$

$$R_{cyl} = \frac{T_1 - T_2}{2\pi L k \frac{T_1 - T_2}{\ln(r_2/r_1)}} = \frac{\ln(r_2/r_1)}{2\pi L k}$$

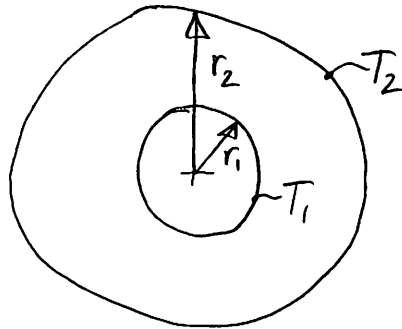
$$\boxed{R_{cyl} = \frac{\ln(r_2/r_1)}{2\pi L k}} \Rightarrow \text{Cylindrical thermal resistance}$$

Note, if $r_2 < r_1$, expression is the same since then $\Delta T = T_2 - T_1$ so negatives would cancel. This is why we typically write:

$$\boxed{R_{cyl} = \frac{|\ln(r_2/r_1)|}{2\pi L k}}$$



③ Sphere



Writing out our spherical coordinate system heat equation:

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = 0$$

Integrate once

$$\int \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = \int 0 \, dr$$

$$r^2 \frac{\partial T}{\partial r} = C_1$$

$$\frac{\partial T}{\partial r} = \frac{C_1}{r^2} \quad (\text{Integrate again})$$

$$\int \partial T = \int \frac{C_1}{r^2} \, dr$$

$$T = -\frac{C_1}{r} + C_2 \Rightarrow \text{Apply boundary cond.}$$

$$T|_{r_1} = T_1 = -\frac{C_1}{r_1} + C_2 \quad (1)$$

$$T|_{r_2} = T_2 = -\frac{C_1}{r_2} + C_2 \quad (2)$$

$$\text{Subtract (2) from (1)} \Rightarrow C_1 = \frac{T_1 - T_2}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)} \Rightarrow \text{Back substitute into (2)}$$

$$C_2 = T_2 + \frac{T_1 - T_2}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)} \cdot \frac{1}{r_2}$$

$$T = \frac{T_1 - T_2}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} \cdot \frac{1}{r} + \frac{T_1 - T_2}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)} \cdot \frac{1}{r_2} + T_2$$

$$\boxed{\frac{T - T_2}{T_1 - T_2} = \frac{\left(\frac{1}{r} - \frac{1}{r_2}\right)}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}} \Rightarrow \text{Spherical temperature profile}$$

Let's try calculating the thermal resistance for the sphere:

$$Q = -kA \left. \frac{\partial T}{\partial r} \right|_r \Rightarrow \frac{\partial T}{\partial r} = \frac{T_1 - T_2}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} \cdot \left(-\frac{1}{r^2}\right)$$

$$A = 4\pi r^2$$

$$Q = k(4\pi r^2) \frac{T_1 - T_2}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} \left(-\frac{1}{r^2}\right) = \boxed{\frac{4\pi k (T_1 - T_2)}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} = Q}$$

But we need $R = \frac{\Delta T}{Q}$

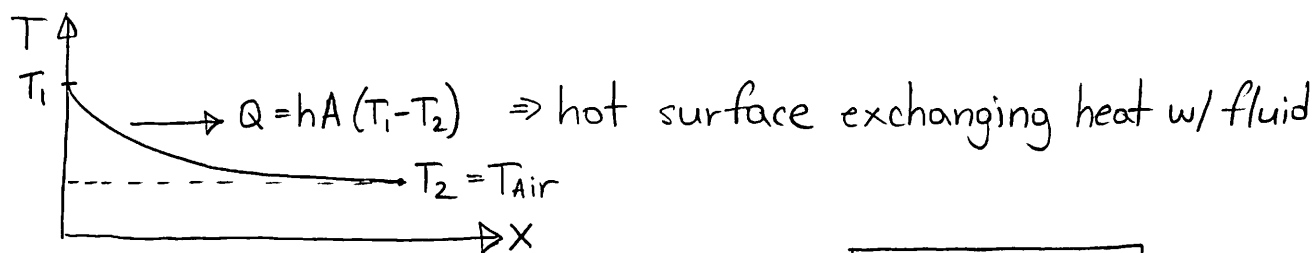
$$\boxed{R_{\text{sph}} = \frac{\left|\frac{1}{r_1} - \frac{1}{r_2}\right|}{4\pi k}}$$

$\Rightarrow$ Spherical thermal resistance
Note, same thing as before with the absolute value rule.

④ Convection thermal resistance (while we're on the topic)

It is often very useful to handle convection the same way. We learned before that: (pg. 15)

$$Q = hA \Delta T ; \quad h \equiv \text{heat transfer coefficient [W/m}^2\cdot\text{K]}$$



$$Q = hA \Delta T \Rightarrow R = \frac{\Delta T}{Q} = \frac{\Delta T}{hA \Delta T} \Rightarrow \boxed{R_{\text{conv}} = \frac{1}{hA}}$$

$\hookrightarrow$ Thermal resistance associated with convection